Ramcharan Rajpurohit

+91-9680103852 | ramcharanrajpurohit@gmail.com | b23ci1032@iitj.ac.in
GitHub | LinkedIn | Website | Codolio

Education

IIT Jodhpur *Sep 2023 – May 2027*
Bachelor of Technology Jodhpur, Rajasthan

Work Experience

- **AI Engineering Intern at PowersMy.biz** *Nov 2025 – Jul 2026*
  - Owned an AI assessment platform from zero to production, now part of the daily study routine of 1,000+ students preparing for JEE, NEET, KCET, and board exams.
  - **Handwritten Evaluation:** engineered a vision-based grader that awards step-wise partial marks against teachers' model solutions and explains each mark, cutting result turnaround from 3 to 4 days of manual checking to under 5 minutes.
  - **Question Generation:** designed a pipeline assembling papers from generative, syllabus, PYQ, and bank sources into competitive, school, or custom patterns, plus a PDF importer (Mistral OCR, DocLayout-YOLO, Gemini) that digitizes full 180-question sets with diagrams intact.
  - **Agentic AI (Amigo):** architected a conversational companion where a LangGraph supervisor routes 5 specialist agents over 18 scoped tools, serving text chat and LiveKit voice in Hindi and English.
  - **Personalised Learning:** created a loop that pinpoints each learner's weak concepts from their rough work, then delivers guided remedy sessions, targeted gap tests, and a tailored crash course.
  - **Apps & Reports:** built the student CBT interface and progress tracking in React and Flutter, plus the teacher app's curriculum hierarchy, test builder, and class analytics for principals.
  - **Backend Performance:** optimized high-concurrency APIs with Redis, atomic MongoDB operations, and off-thread bcrypt to sustain peak load.
  - **Distributed Systems:** split evaluation into an asynchronous queue across 2 services that returns objective scores instantly while subjective responses are processed in the background.
  - **Cloud & DevOps:** led GCP deployment on Cloud Run with a VPC connector, autoscaling, and concurrency tuning to absorb traffic spikes, shipped through GitHub Actions pipelines.

Projects

- **DyPol:** Engineering Activity & Insights **[Live Site]** | **[GitHub]**
  - Built a platform to track engineering activity across teams and projects, generating weekly reports, contributor insights, PR tracking, and stalled-work alerts from GitHub data.
  - Developed a natural-language engineering assistant using LangGraph orchestration, specialized agents, vector embeddings with Chroma and MongoDB, and GitHub REST and GraphQL APIs, with tool scoping, delegation limits, and cost tracking.
- **DSA Visualizer:** Interactive Algorithm Learning Platform **[Live Site]** | **[GitHub]**
  - Developed a framework-free algorithm visualizer in raw ES6 and HTML5 Canvas, animating 15+ sorting, graph, and tree algorithms step by step.
  - Architected a class-based OOP rendering engine where each algorithm is a self-contained plug-in subclass, with frame-by-frame controls for recursion tracing and backward stepping.

Technical Skills

- **AI, ML & GenAI:** LangGraph, LangChain, Multi-Agent Systems, Gemini API, OpenAI API, Vision Models, Realtime Voice Agents (LiveKit, STT, TTS), Prompt Engineering
- **Programming Languages:** C++, C, Python, JavaScript, TypeScript
- **Frontend Development:** React.js, HTML5, CSS3, TailwindCSS, Redux, Flutter
- **Backend Development:** Node.js, Express.js, FastAPI, REST APIs, GraphQL, SSE Streaming, WebRTC, JWT, OAuth 2.0, MongoDB, Redis
- **Cloud & Tools:** GCP (Cloud Run, VPC), Docker, GitHub Actions, Git, Postman, Linux, Vercel

Achievements

- Authored a **Linux kernel** patch merged into mainline, fixing a broken mic-mute LED on a personal HP laptop after root-causing the missing audio driver quirk. **[Commit]**
- Ranked **1st of 150** in IIT Jodhpur's AI Chess Tournament, unbeaten in 300 games as white and black.
- Achieved **2nd District Rank** in Class 12 RBSE Board Exams with a 98.20% aggregate.

Leadership

- As Core Member of the Dramatics Society (**Dramebazz**), IIT Jodhpur, coordinated events and creative direction, and directed stage productions at **Inter IIT Cultural Meet 7.0**.